

4. Non-REM sleep may be associated with:

- (a) Sleep-talking
- (b) Sleep-walking (somnambulism)
- (c) Bed wetting (nocturnal enuresis)

5. REM sleep may be associated with:

- (a) Excessive swallowing
- (b) Teeth grinding (bruxism)
- (c) REM behavior disorder: no ↓ in ms. tone

6. Sleep apnea

Lecture (16): Functions of the different parts of the eye (I)

(Dr. Wagdy Nashaat Habeeb)

Eye Ball

➤ Structure of the eye ball:

(A) 3 layers:

1. Outer fibrous protective layer:

- a) Sclera: post. 5/6, opaque
- b) Cornea: ant. 1/6, transparent

2. Middle vascular layer (Uveal tract):

- a) Choroid: post. 5/6, pigmented, contains BVs
- b) Ciliary body: ant. ring, contains ciliary ms., ciliary processes & suspensory ligaments (SL)

c) Iris: colored portion, perforated (pupil)

3. Inner nervous layer:

Retina: lines the post. 2/3 of choroid, 10 layers, contains the photoreceptors (rods & cones), with: *Macula lutea & fovea centralis *Optic disc

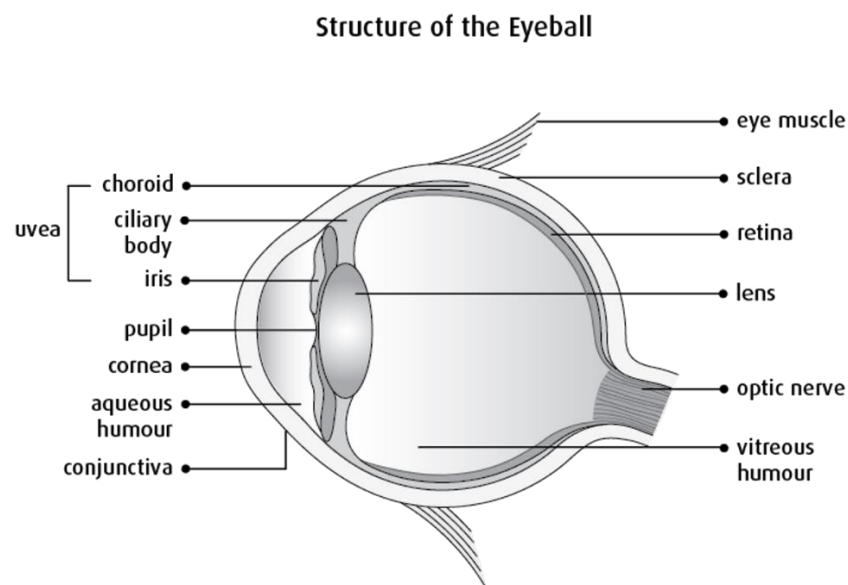
(B) 3 contents:

1. Lens: attached to CB by SL

2. Aqueous humour (AH): ant. to the lens (fills the ant. & post. chambers)

3. Vitreous humour (VH): post. to the lens

N.B. 2 & 3 represent the fluid system of eye



1. Aqueous Humour (AH)

➤ Definition:

Clear watery fluid that fills the space between lens & cornea, *i.e.*, the ant. & post. chambers

➤ Characters:

1. pH: alkaline

2. Volume: 0.3 ml

3. Rate of formation & drainage: 3 $\mu\text{l}/\text{min}$

➤ Secretion:

From inner non-pigmented epithelium of ciliary processes of the ciliary body

➤ Mechanism of formation:

1. Active secretion of Na^+ by Na^+-K^+ pump, followed by passive transport of:

* Cl^- & HCO_3^- : by diffusion (*N.B.* HCO_3^- is formed by carbonic anhydrase enzyme)

* H_2O : by osmosis

2. Nutrients (glucose, amino acids & ascorbic acid) are transported by active transport or facilitated diffusion

➤ Circulation & Drainage:

Inflow

Outflow

!——# Ciliary processes → Post. chamber → Pupil → Ant. Chamber !——#
Irido-corneal angle (ICA) → Trabecular meshwork (spaces of Fontana) →
Schlemm's canal → Episcleral veins

N.B. AH moves by the following mechanisms:

1. Intermittent variation in the intraocular pressure (IOP) which is produced by variations in ABP during cardiac & respiratory cycles
2. The continuous movements of the eyelids
3. Convection currents in AH as the temperature of the iris is 3-5 °C higher than that of the cornea

➤ Functions:

1. It supplies O₂ & nutrients to the avascular structure (cornea & lens)
2. It drains metabolites from cornea & lens
3. It maintains IOP → keep the spherical shape of eye
4. It is one of the refractive media of the eye

➤ Intraocular Pressure (IOP):

*Definition: Pr. inside the eyeball

*Normal value: 12 – 20 mmHg

- It depends mainly on AH volume
- It shows diurnal variation (highest in the morning & lowest in the evening)
- It falls to about 8 mmHg after death due to arrest of the ocular circulation
- It is maintained nearly equal to the choroidal capillary BP. If the later increases, e.g., by coughing or straining, IOP also increases and *vice versa*
- It is kept fairly constant in spite of the pulsatile & respiratory variations in ABP

*Measurement: Tonometry, either by:

- Indentation method “Schiotz tonometer”, which is applied to the cornea after anesthetizing it by cocaine drops
- Digital applanation method “air puff tonometer or Tono-Pen”

*Regulation: via AH drainage (How?)

*Functions:

1. It maintains the spherical shape of eye
2. It is important for the normal focusing mechanism of eye (How?)

➤ Glaucoma:

*Definition: Optic nerve damage, often associated with $\uparrow\uparrow$ IOP > 21 mmHg

*Cause: $\uparrow\uparrow$ resistance of AH outflow

*Types: 2 major types:

Primary open-angle glaucoma (POAG)	Closed-angle glaucoma (CAG)
The most common	Rare
Chronic (a lifelong condition)	Acute (an emergency condition)

ICA: open & wide (the cause is slow obstruction of the draining canals beyond ICA)	ICA: closed or narrow ($\rightarrow$ sudden $\uparrow\uparrow$ in IOP)
Symptoms: usually not noticed	Symptoms: severe, very noticeable

N.B. Other types include normal (or low) tension glaucoma, congenital glaucoma & 2ry glaucoma

*C/P:

1. Blindness (gradual painless loss of vision in POAG): due to $\uparrow\uparrow$ IOP $\rightarrow$ compression of optic nerve fibers & retinal artery $\rightarrow$ retinal atrophy
2. Severe eye pain & headache (in case of CAG)
3. Mydriasis
4. Failure of accommodation to near vision (Why?)

*ttt:

1. Medical:

- (a) $\downarrow$ AH formation: Carbonic anhydrase inhibitor (Diamox)
- (b) $\uparrow$ AH drainage: Parasympathomimetics (eserine or pilocarpine)

2. Surgical: widening of trabecular meshwork by microsurgery or laser

2. Vitreous Humour

➤ Definition:

Clear gelatinous fluid that fills the space between lens & retina

➤ Functions:

1. It supports lens & retina in their position
2. It supplies lens & retina with glucose
3. It maintains the spherical shape of eye
4. It is one of the refractive media of the eye
5. Shock absorber → protect retina from injury

3. Cornea

➤ Functions:

1. Allows entry of light rays
2. The main refractive medium of eye (Why?)

N.B. Its regular curvature leads to formation of sharp retinal images

3. Protection:

- (a) *With sclera*: protect inner delicate structures
- (b) *With lens*: protect retina from UV rays
- (c) *Corneal reflex*: protect eye from harmful FB

4. It is permeable to isotonic fluids → allow various eye drops to get easily into the eye

➤ Protection of cornea:

1. Closure of eyelids:

- (a) Blinking: spontaneous, reflex or voluntary
- (b) Blepharospasm: in eye injury

2. Tear film:

- (a) Optical: it maintains uniform corneal surface
- (b) Nutritive: it allows gas exchange (.) air & cornea
- (c) Lubricant: it facilitates lid movement
- (d) Protective: it contains lysozymes, lactoferrin, IgA

3. Corneal reflex

➤ Corneal reflex:

*Stimulus: Touching cornea of one eye by a foreign body, *e.g.*, a piece of cotton

*Receptor: Unmyelinated free nerve endings

*Afferent: Ophthalmic division of 5th CN to trigeminal nucleus in pons

*Center: Sup. colliculus in medulla

*Efferent: 7th CN on both sides

*Response: Bilateral blinking (cont. of orbicularis oculi)

*Significance:

- 1. Protection
- 2. Tests the integrity of 5th & 7th CN (How?)
- 3. Staging of anesthesia (How?)

➤ Causes of corneal transparency:

1. Avascular (Q. How does the cornea receive its O₂ supply & nutrients?)

N.B. Vitamin B2 (riboflavin) ↓↓ → corneal vascularization → loss of corneal transparency

2. Unmyelinated nerve endings

3. Regular arrangement of corneal epithelium & collagen fibers

4. Relative corneal dehydration: *via*

(a) Metabolic pump: of Na⁺ from cornea to AH by Na⁺-K⁺ pump

(b) Osmosis: H₂O follows Na⁺ to AH & tear film

N.B. Vitamin A ↓↓ → xerophthalmia (dry eye) → impairment of corneal transparency

4. Sclera

➤ Functions:

1. Protection of inner delicate structures

2. Gives attachment to extraocular ms. (EOM)

Lecture (17): Functions of the different parts of the eye (II)

(Dr. Wagdy Nashaat Habeeb)

5. Crystalline Lens

➤ Structure & Properties:

1. Biconvex transparent structure behind the iris (its post. surface is more convex than the ant. surface during rest)

2. Enclosed in an elastic capsule, but it becomes progressively hard with